

Caroline “CC” LeBlanc

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EDUCATION

Master of Business Administration, May 2027

The University of Alabama, Manderson Graduate School of Business, Tuscaloosa, AL

GPA: 3.75/4.0 Focus Area of Study: *Business Analytics*

Bachelor of Science in Mechanical Engineering, May 2026

The University of Alabama, Lee J. Styslinger College of Engineering, Tuscaloosa, AL

GPA: 3.49/4.0 Honors College *Fundamentals of Engineering (FE) Exam – Passed*

SKILLS

MATLAB, Python, Power BI, Microsoft Excel, Motec i2, SolidWorks, Simulink, AutoCAD, Bosch WinDarab (Beginner), Cadmatic, WinGeo3, Adobe Creative Cloud, TIG Welding

WORK EXPERIENCE

Engineering Intern

May 2026 – August 2026

BF Goodrich / Michelin, Tuscaloosa, AL

- Implemented a profilometer measurement system on treadline operations, bringing tread cut angle from 83° to the ~90° specification and raising process capability (Cp) from 1.02 to 1.12 after the first upgrade
- Designed new skiver blade components in SolidWorks to improve tread cut quality and reduce variability along treadline systems
- Conducted flow analysis of spray bath systems in tread manufacturing lines to identify process optimization opportunities
- Gained hands-on knowledge of tire manufacturing processes across carcass building and tread extrusion processes

Engineering Intern

May 2025 – August 2025

Heinen & Hopman, Spakenburg, The Netherlands

- Performed heat transmission calculations in company software to cross-check Excel-based hand calculations
- Produced AutoCAD ventilation and HVAC system drawings for marine applications

RELEVANT EXPERIENCE

Vehicle Dynamics Senior Engineer, Suspension Assistant Lead

May 2024 – May 2026

Alabama FSAE (Crimson Racing), The University of Alabama, Tuscaloosa, AL

- Diagnosed a rear-axle oscillation at the car's natural frequency and retuned the damper setup, cutting lateral-g oscillation amplitude by 56% and helping the team place 6th and 8th overall at FSAE Michigan in 2025 and 2026
- Redesigned suspension bumpstops from SBR rubber to a 3D-printed TPU part (54 samples tested), cutting manufacturing deviation by 83% (192.7 to 32.5 lbf/in), and corrected a tie rod-to-clevis angle from 14°, beyond the manufacturer's 10° maximum, down to the recommended 5°
- Designed the suspension geometry package and performance targets for camber, toe, and caster across the full travel range, then used a FARO arm to confirm motion ratios stayed within 1.3% of design
- Developed a Pacejka-based Magic Formula tire model in MATLAB and applied it to the team's tire decision matrix, cutting tire selection decision time by an estimated 50%

Communications Lead

May 2025 – present

Alabama FSAE (Crimson Racing), The University of Alabama, Tuscaloosa, AL

- Grew team social media following from 2,000 to 3,500 (+75%) and expanded active subteams from 1 to 7 in one year
- Lead a subteam of 4, providing mentorship in Adobe software to ensure sustainable knowledge transfer among members
- Design and manage all digital and print communications, including social media content, website updates, and university outreach materials such as flyers and promotional graphics

EDAG Student Team Lead

August 2025 – May 2026

STEM GBA Capstone Project, The University of Alabama, Tuscaloosa, AL

- Organized timelines and project scope objectives for a student team of four
- Coordinated meetings and communicated final products to German clients to deliver a manufacturing facilities guideline

LEADERSHIP & ACTIVITIES

Alabama FSAE (Crimson Racing)

Alabama Collegiate Recovery Community

Society of Automotive Engineers